ST115
Lab-2

Dr. Elijah Meyer & Nick Mann

Goals for today

- Practice with R
- Practice with Quarto
- Be able to create a PDF + submit on Gradescope

As a reminder:

- Attendance is graded (I will be passing a sheet around)
- Lab assignments are graded (on Gradescope)

Warm Up

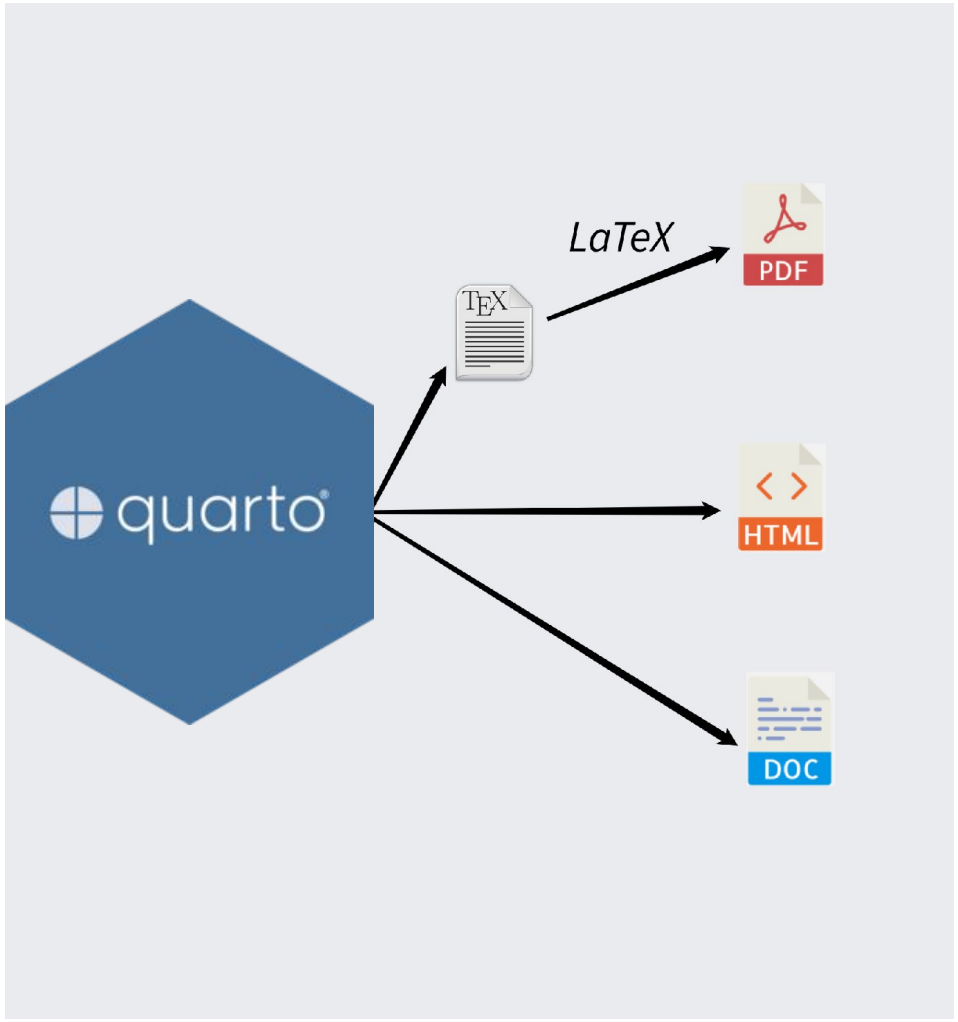
- What is the difference between a
  - .R file
  - .qmd file

And when would we consider using each?

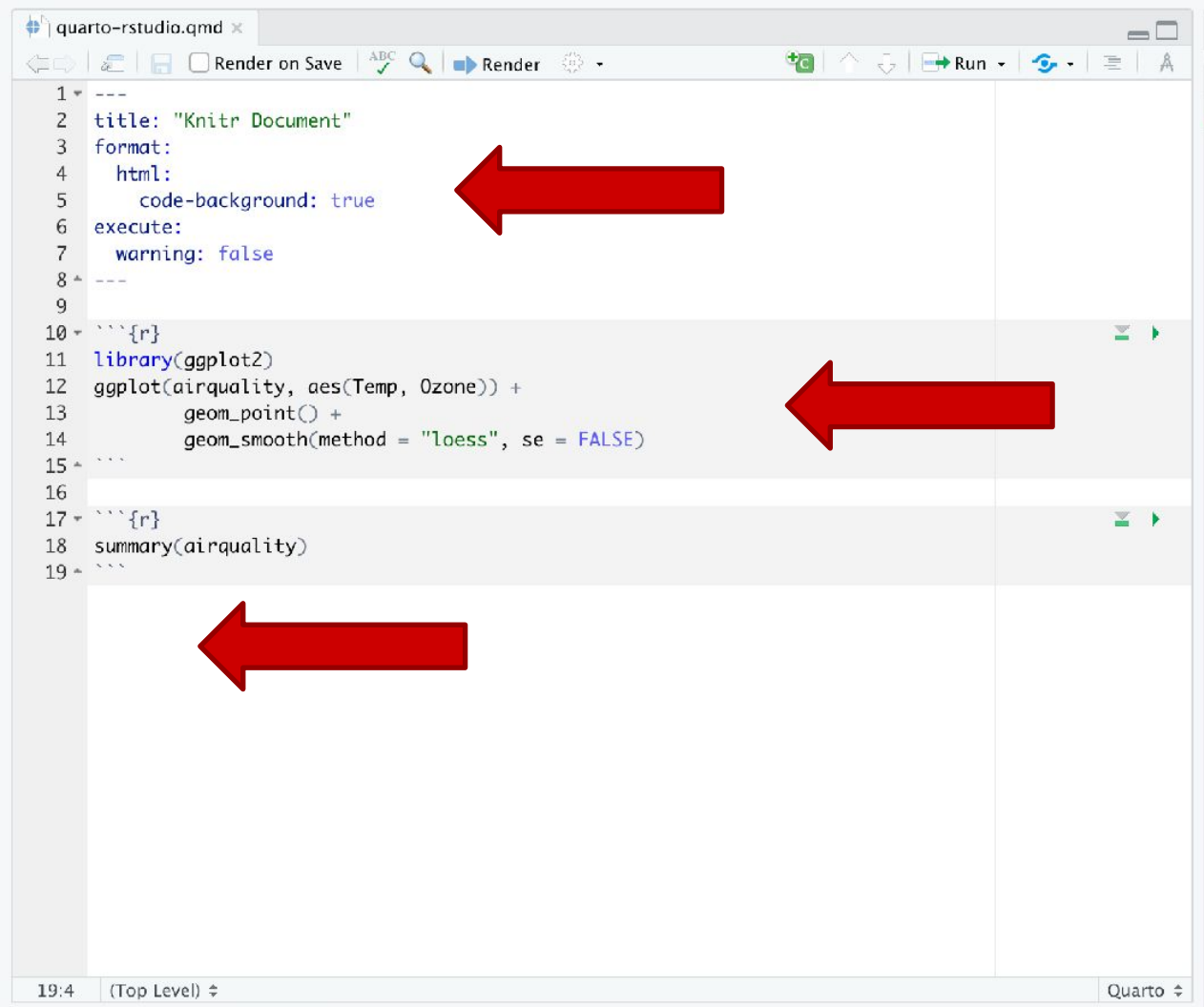
Warm Up

File	What it is	Typical use
.R	An R script	Writing and running R code (scratch paper)
.qmd	A Quarto document	Combining text, R code, results, tables, graphs, etc. into a report/presentation

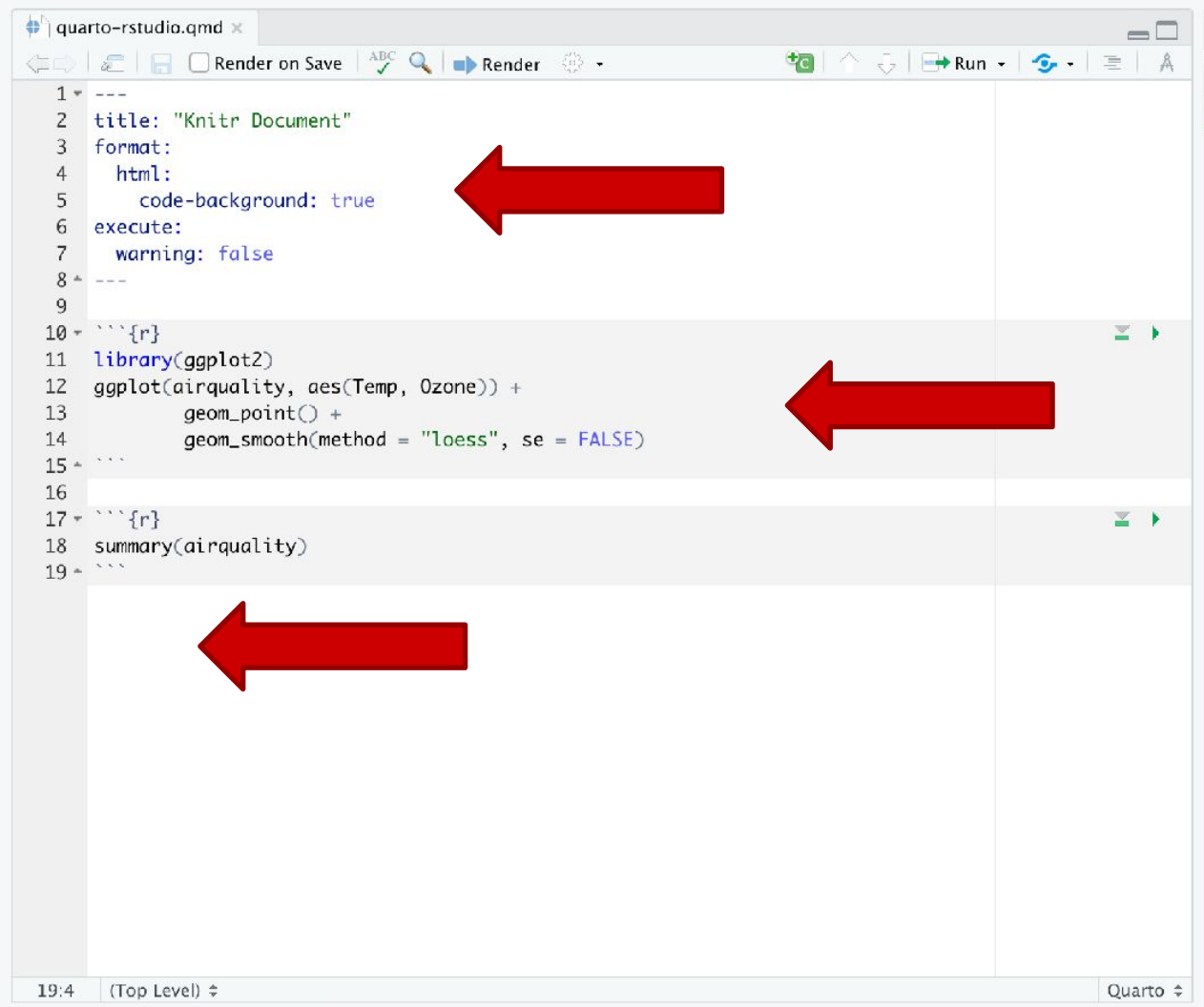
Quarto reports



Parts of a Quarto Document



Parts of a Quarto Document



YAML – information that goes at the top of your document + tells Quarto what to render

Code chunk – where we write code to compile

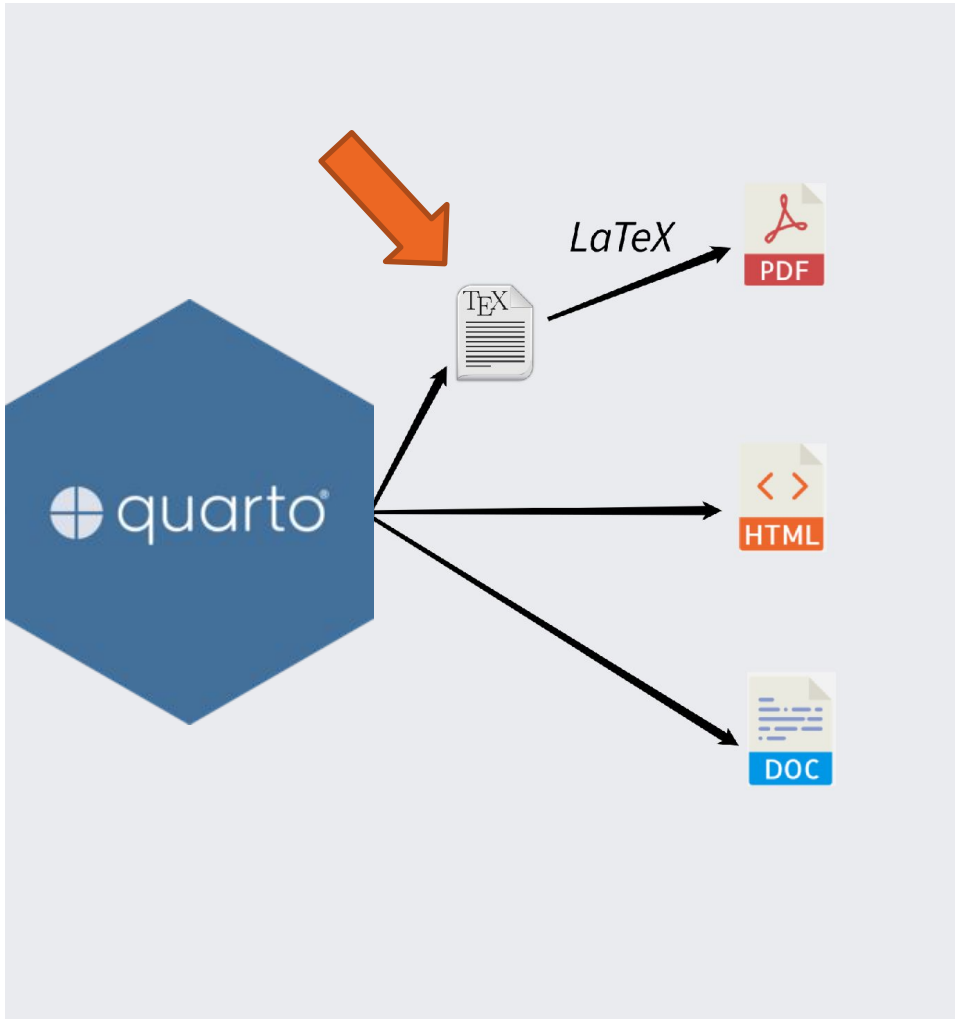
Text– where we write text

Setting up YAML

Dr. Meyer's lecture slides (slide linked and the next one):
https://st115-fall26.github.io/class/slides/03-Quarto.html#/yaml-1

Which Example is correct?

We need to install this!



tinytex

In your console:

- install.packages("tinytex")
- tinytex::install_tinytex()

We will learn more about packages soon! For now, think of this as a way for us to get the necessary software to compile PDFs.

If you are on mac, you should already have this installed, and can skip this step!

Lab Assessment

Lab Outline

1. Problems 1 & 2 - Demo
    a. Questions?

Lab Outline

1. Problems 1 & 2 - Demo
    a. Questions
2. Work through problems 3-5 on your own, try to render

Lab Outline

1. Problems 1 & 2 - Demo
   a. Questions
2. Work through problems 3-5 on your own, try to render
   a. Solution to 3
   b. Answers to problems 4 & 5?

````
```{r}
b <- c(1, 2, 3, 4, 5)
b
```
````

```
[1] 1 2 3 4 5
```

Lab Outline

1. Problems 1 & 2 - Demo
   a. Questions
2. Work through problems 3-5 on your own, try to render
   a. Solution to 3
   b. Answers to problems 4 & 5?
3. Finish problems 6-8 on your own

````
```{r}
b <- c(1, 2, 3, 4, 5)
b
```
````

```
[1] 1 2 3 4 5
```

Lab Outline

1. Problems 1 & 2 - Demo
   a. Questions
2. Work through problems 3-5 on your own, try to render
   a. Solution to 3
   b. Answers to problems 4 & 5?
3. Finish problems 6-8 on your own
   a. Answers to problems 7 & 8?

````
```{r}
b <- c(1, 2, 3, 4, 5)
b
```
````

```
[1] 1 2 3 4 5
```

Lab Outline

1. Problems 1 & 2 - Demo
   a. Questions
2. Work through problems 3-5 on your own, try to render
   a. Solution to 3
   b. Answers to problems 4 & 5?
3. Finish problems 6-8 on your own
   a. Answers to problems 7 & 8?
4. Install tinytex!

````
```{r}
b <- c(1, 2, 3, 4, 5)
b
```
````

```
[1] 1 2 3 4 5
```

pdf

Next, change your format to pdf in your YAML, and click render!

format: pdf

And turn this in on Gradescope! Let's demonstrate.